[METRIC]
A-A-52549
March 15, 1996
SUPERSEDING
MIL-N-62623(AT)
30 March 1988

COMMERCIAL ITEM DESCRIPTION

NICKEL BASE ALLOY, SHEET, CORROSION- AND HEAT-RESISTANT (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers sheet stock of a corrosion- and heat-resistant nickel base alloy, referred to herein as "alloy". This alloy is intended primarily for parts requiring good corrosion- and oxidation-resistance at temperatures up to approximately 1093 degrees Celsius (°C). Alloy may be used where such parts require welding.

2. SALIENT CHARACTERISTICS

2.1 Materials. Materials shall be uniform in quality and condition, clean, sound, and free from foreign materials and internal and external imperfections detrimental to fabrication or performance of parts. Conformance to these requirements shall be established by visual inspection for surface quality of the final product and metallographic examination of the structure. The use of recovered materials made in compliance with regulatory requirements is acceptable provided that all requirements of this CID are met (see 3.1).

2.1.1 Delivery condition. Unless otherwise specified (see 6.2), the alloy shall be delivered cold rolled and bright annealed in a protective atmosphere, such as hydrogen. The use of dissociated ammonia or N2/H2 mixtures is prohibited. The product may be leveled after the last anneal to enhance flatness of the product.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 9630

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2.1.2 Chemical composition. The alloy shall be nickel-chrome, solid solution strengthened, Unified Numbering System (UNS) N06625, in the form of sheet stock. The chemical composition of the alloy shall conform to the percentage by weight of each element as specified in ASTM B443 or equivalent. Determination of cobalt is not required for routine acceptance.

2.2 Design and construction.

2.2.1 Thickness. Alloy shall be not greater than 0.51 millimeter (mm) thick.

2.2.2 Grain size. The grain size of the alloy shall be ASTM Micro-Grain Size #5 or finer in accordance with ASTM E112.

2.2.3 Surface discoloration. Unless otherwise specified (see 6.2), there shall be no evidence of surface discoloration present on the alloy.

2.3 Performance.

2.3.1 Tensile properties. The tensile properties of the alloy shall be as follows when the alloy is tested in accordance with ASTM E8:

a. Ultimate strength: Not less than 724 megapascals (MPa)
b. Yield strength at 0.2 percent (%) offset: 331 to 393 MPa
c. Elongation in 50.8 mm: Not less than 35%

2.3.2 Stress rupture. Alloy shall not fail in less than 23 hours and elongation shall be not less than 15% in 50.8 mm when the alloy is subjected to a continuous axial load of 114 MPa in accordance with ASTM E139 at a temperature of 816 $\pm$ 3°C.

2.3.3 Bending. Alloy shall withstand, without cracking, bending at room temperature through an angle of 180 degrees around a diameter equal to the nominal thickness of the alloy, with the axis of the bend parallel to the direction of rolling.

2.4 Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), the heat number, the alloy nominal thickness, and the part identification number (PIN) (see 6.3). When identification marking is performed directly on the alloy, the marking fluid shall have no deleterious effect on the alloy and shall be sufficiently stable to withstand normal handling. The marking fluid shall be capable of being removed in a hot alkaline solution. Unless otherwise specified (see 6.2), the alloy shall be marked on one face as follows:

a. The rows of characters shall be spaced not over 914 mm apart.
b. The characters shall be of a size to be clearly legible.

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3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government documents. Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

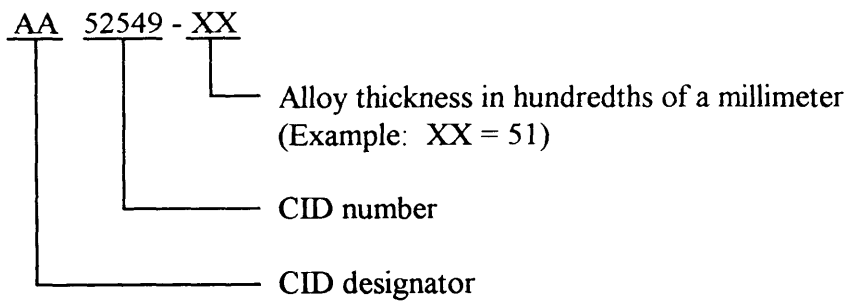
6.1.2 Non-Government documents. Copies of SAE HS 1086 "Metals and Alloys in the Unified Numbering System (UNS)" are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096. Copies of ASTM B443 "Standard Specification for Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) Plate, Sheet, and Strip", ASTM E8 "Standard Test Methods for Tension Testing of Metallic Materials", ASTM E112 "Standard Test Methods for Determining Average Grain Size", and ASTM E139 "Standard Practice for Conducting Creep, Creep-Rupture, and Stress-Rupture Tests of Metallic Materials" are available from the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

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6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Size of alloy (length, width, thickness) and PIN number.
d. If the alloy should be delivered in a condition other than as specified.
e. If surface discoloration should be other than as specified.
f. If marking should be other than as specified.
g. Selection of applicable level and packaging requirements.

6.3 Part identification number (PIN). The PINs to be used for the alloys acquired by this CID are created as follows:



6.4 Cross-reference. Alloys conforming to this CID are interchangeable/substitutable with alloys conforming to MIL-N-62623(AT), dated 30 March 1988.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army - AT

(Project 9630-0035)

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